Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Advanced Robotics & Manufacturing: Digital Work Instructions** "AI-powered "Instant Instructions" for ultra-rapid creation and deployment of digital work instructions on the shop floor"
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power (√), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )

Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		*Mechanical · Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	RFID/USN	*Service Platforms and Application Software*	*Robotics/Automation Machinery*	*Other Robotics/Automation Machinery Software*
Concept of Proposed Technology	Instant Instructions enables any frontline worker to record task execution on a tablet or phone; an AI pipeline then converts that footage into step-by-step, text-plus-video digital work instructions that can be quality-checked, improved, translated into multiple languages, and deployed securely inside a company environment or integrated with existing digital work instruction platforms. The result is fast, accurate, and scalable documentation produced by the people who actually perform the work, not only by engineers. Applicability spans manufacturing, maintenance, production, and healthcare.			
Rationale for Joint R&D Project	SMEs are advancing in smart manufacturing and digitalization yet face constraints: rapid onboarding needs, aging workforce knowledge drain, variability in process quality, and high documentation costs. Joint R&D with TNO will adapt and validate Instant Instructions to industrial contexts, domain-specific templates (electronics, precision machining, medical devices), on-premise or sovereign cloud options, and integration with Korean MES/QMS platforms. The collaboration accelerates localization, standard compliance, and field validation with pilot factories in Korea, enabling exportable solutions for the broader APAC market.			
Key Words	1. Digital Work Instructions (DWI) 2. Knowledge Capture / Operator Guidance 3. AI-generated SOPs / Multilingual Instructions 4. Smart Factory / Industry 4.0 Enablement 5. Manufacturing Quality & Onboarding Acceleration			

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	**Start of the Project**	TRL 4 (Lab validation of integrated modules)	**End of the Project**	TRL 6 (Pilot demonstration in SME environment: Production & performance evaluation of prototypes at a pilot scale)
R&D Objectives	**Key R&D objectives:** **1.** Instant Instructions for the Korean language, including speech recognition, terminology, and user interface. **2.** Develop sector-specific templates for electronics assembly, precision machining, secondary battery components, and medical devices. **3.** Integrate the platform with commonly used Korean MES, QMS, and document management systems; include optional AR/VR guidance modules. **4.** Enhance platform security, role-based access control, compliance features, and on-premise/cloud flexibility suitable for Korean SMEs. **5.** Test multi source (video, CAD, documents, etc.) instruction development. **6.** Conduct multi-site pilot projects in Korea to evaluate performance, refine functionality, and prepare commercialization guidelines. . The overarching R&D objective of this joint program is to evolve TNO's "Instant Instructions"			

	technology into a next-generation Instant Digital Work Instruction platform that enables highly accurate, context-aware, and rapidly generated work instructions tailored to the needs of Korean SMEs in advanced manufacturing sectors. Building on the insights, architectures, and human-centric design principles developed in previous programs, the project aims to combine automated knowledge capture, multimodal sensing, semantic interpretation, and AI-driven instruction generation into an integrated solution capable of supporting *zero-defect, zero-delay, and zero-training* ambitions for high-mix/low-volume production environments. A core objective is to enhance the technology stack with advanced AI capabilities such as multimodal reasoning, speech-to-process mapping, CAD-to-instruction translation, and automated error-detection while ensuring these augmentations remain fully aligned with the operational realities, quality frameworks, and regulatory requirements of Korean factories. This includes extending the platform to incorporate multi-source inputs video, CAD, sensor data, tribal knowledge, SOPs, enabling the system to generate highly accurate, step-wise, and domain-specific procedures for electronics assembly, precision machining, medical devices manufacturing, and next-generation semiconductor processes. Furthermore, the project will strengthen the human-centric dimension of the system. Following the principles of the **Workplace Information Space (WIS)** and **Digital Employee Passport (DEP)**, the platform will support personalized instruction delivery, skill-adaptive guidance levels, and advanced feedback loops that allow SMEs to retain knowledge, accelerate onboarding, and improve well-being, inclusiveness, and operator safety. Finally—and critically—the **R&D objectives will be further detailed and prioritized based on the concrete use cases, factory conditions, and process requirements of the Korean SME partners**. Their input will drive localization, domain templating, UI/UX adaptation, security requirements, and integration pathways with Korean MES/QMS/ERP platforms.
R&D Target Specifications	**Key R&D specifications:** 1. Reduce time required to create standardized work instructions by at least 70%. 2. Reduce operator onboarding time by at least 30% in pilot environments. 3. Improve first-time-right and reduce process errors by at least 25%. 4. Deliver high-quality Korean–English bilingual instructions with domain-specific glossaries. 5. Successfully integrate with at least two Korean MES/QMS systems and one AR-based instruction viewer. The technical and functional targets of the Digital Work Instructions (DWI) system are designed to demonstrate measurable performance improvements in instruction accuracy, generation speed, operator autonomy, and quality assurance. Drawing from the Minimum Viable Product for DWI and the operational benchmarks of Korean manufacturing SMEs, the project will deliver a fully operational prototype that meets or exceeds the following targets: 1. **Ultra-fast work instruction generation** • Achieve a minimum 70–80% reduction in time required to create, validate, and deploy standardized work instructions through automated AI-generated procedures based on multimodal input streams like video, CAD, legacy SOPs, sensor data, etc. • Support near-real-time regeneration of instructions after engineering changes, process updates, or detected quality deviations.

	2. Operator-centric performance improvement • Reduce onboarding and training time by ≥30–40%, using adaptive instruction levels, personalized guidance, and context-aware feedback rooted in DEP-like worker profiles. • Decrease cognitive load and variability in task execution by leveraging WIS-based augmentation methods aligned with Industry 5.0 human-centric design principles. **3. Quality, reliability, and zero-defect contribution** • Achieve ≥25% reduction in first-time-right errors through consistent, AI-verified instructions and optional AR-assisted validation. • Integrate anomaly-detection and automated checklist updates to prevent propagation of recurring defects. **4. Localization and bilingual technical accuracy** • Deliver high-fidelity Korean–English bilingual work instructions, enriched with sector-specific terminology (electronics, machining, medical devices, aerospace) and industry-approved controlled vocabulary. **5. Systems integration and interoperability** • Successfully interface with at least two Korean MES/QMS/ERP systems, leveraging OPC UA, AAS, or equivalent standardized interfaces, ensuring secure and sovereign data flow. • Integrate at least one AR-based viewing system for real-time projection- or wearables-based operator support. **6. Security, deployment flexibility, and SME-readiness** • Support on-premise, hybrid, and sovereign-cloud deployments aligned with Korean cybersecurity standards. • Deliver role-based access control, lineage tracking, auditability, and full version control of instructions. As with the objectives, the final target specifications will be refined in collaboration with Korean SMEs, ensuring that the performance indicators address their real production bottlenecks, workforce skill profiles, data environments, and business constraints. The system will be iteratively evaluated across multiple Korean pilot sites to ensure technical fit, usability, and industrial relevance.
R&D Plan	**High level planning to be adjusted based on the use cases of Korean SMEs:** • **Month 0-6:** Define User Requirement Specification and System requirement specification based on SME use-cases for DWI. • **Month 6-18:** develop and extend existing lab prototype DWI • **Month 18-24:** Pilot demonstration, performance validation, • **Month > 24:** technology & commercialization roadmap development and industrial implementation at participating SME's

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Provide the core Instant Instructions technology platform, AI models, and reference architectures. • Lead localization research and development, including speech recognition, translation, and workflow design. • Oversee pilot execution, KPI design, result analysis, and joint commercialization preparation. • Provide deep expertise in human-centred design, industrial digitalization, and operator support systems.
Expected R&R of Korean SME Partner	• Provide factory / test environments, subject-matter experts, and operational data for localization and testing. • Lead integration into Korean MES/QMS platforms and support UI/UX localization. • Conduct on-site trials, gather operator feedback, measure KPIs, and support refinement. • Prepare commercialization and service models for the Korean market.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• Delivery of a (preliminary) Korea-localized digital work instruction platform tailored to SME needs. • Significant improvements in training speed, process consistency, and error reduction in pilot factories. • Direct cost savings through reduced documentation effort and improved operational efficiency. • Strengthening of Korea's strategic competency in smart manufacturing, AI-assisted operator support, and digital knowledge retention. • Establishment of a scalable solution suitable for multiple Korean sectors and future regional expansion.

6. References

TNO DWI: https://ventures.tno.nl/portfolio/instant-instructions/

Publication: Jobs and Skills in Smart Manufacturing SMEs: This research (incl. TNO Paul Preenen and in cooperation with UTwente(Sebastian Thiede) explores how production workers can adopt smart technologies (AI, AR/VR, robotics) and aligns with the Industry 5.0 vision. nijland-2025-jobs.pdf

Publication: Capturing and transferring knowledge through AI generated digital work instructions. STS conference in Trondheim (SocialTechnical Systems) September 2025